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Cyber-Security Awareness Report

Mini Project 4 · Team Report

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Declaration

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Academic integrity and use of Generative AI (GAI)

We have worked closely with other members of the team and ourselves to ensure, to the best of our ability, that each section of this report was written by the individual responsible and that they have responsible referenced any sources from which they got their information. GAI was used in certain cases but with justifiable reasons.

GAI tools used: Google Gemini and ChatGPT

How they were used: GAI was used to assist with the introduction and conclusion statements. GAI was also used in certain sections to tidy up the look and feel of the section lead's writing. However, all sections are written by the person responsible, to the best of our knowledge, and referenced appropriately.

Contents

1. Introduction 2
   1.1 Purpose and Importance 2
   1.2 Scope and Intended Audience 2
2. Password Hygiene 3
   2.1 Introduction 3
   2.2 Creating Strong Passwords 3
   2.3 Avoiding Password Reuse 3
   2.4 Using Password Managers 4
   2.5 Multi-Factor Authentication 5
   2.6 Storing and Protecting Passwords 5
   2.7 Conclusion 6

3. Two-Factor and Multi-Factor Authentication 6
3.1 Introduction 6
3.2 How Two-Factor Authentication Works 6
3.3 Benefits of Two-Factor Authentication 7
3.4 Types of Two-Factor Authentication 7
3.5 Limitations and Security Risks 8
3.6 Best Practices for Using Two-Factor Authentication 9
3.7 Conclusion 9
4. Phishing and Social Engineering 9
4.1 Introduction to Phishing 9
4.2 Why is phishing so effective? 10
4.3 How to recognise a phishing attack 11
4.4 What should you do if you suspect you're a victim of phishing? 12
5. Safe Use of Devices 12
5.1 Introduction 12
5.2 Keeping Devices and Software Updated 12
5.3 Using Strong Device Passwords and Authentication 13
5.4 Protection Against Malware 13
5.5 Physical Security of Devices 14
5.6 Safe Use of Smart Devices 14
5.7 Backing Up Important Data 15
5.8 Conclusion 15
6. Safe Use of Public Wi-Fi 16
6.1 Introduction to Public Wi-Fi Security 16
6.2 The Evolution of Web Encryption and False Trust 16
6.3 Human Factors: Psychology, Trust, and Decision Making 17
6.4 Technical Threats: Interception and Exploitation 17
6.5 Mitigation Strategies and Security Best Practices 18
7. Conclusion 19

8. References 19
Appendix A: Team Contribution Record 22
Appendix B: Individual Source Log 22

1. Introduction

Section lead: Mitchell **Peer reviewed by:** Kenz

1.1 Purpose and Importance

Cybersecurity is no longer strictly an IT responsibility—it is an operational priority shared by every individual across an organization. As digital environments expand and modern threat vectors grow increasingly sophisticated, human behavior remains one of the primary targets for cybercriminals. Standard technical defenses, such as firewalls and network filters, are only as effective as the daily security habits of the personnel operating within the network.

The primary purpose of this report is to establish a proactive security posture by bridging the gap between technological safeguards and user behavior. By addressing five critical threat vectors—**phishing attacks**, **weak password hygiene**, **inadequate multi-factor authentication (2FA)**, **unsecured physical and smart devices**, and **unprotected public Wi-Fi networks**—this report equips individuals with the knowledge to recognize, prevent, and mitigate common cyber threats. Fostering a security-conscious mindset significantly reduces the risk of data breaches, credential theft, financial losses, and operational disruption.

1.2 Scope and Intended Audience

Scope

This report provides practical guidelines, threat analysis, and actionable best practices across five core domains:

- **Phishing & Social Engineering:** Spotting deceptive communications, fake URLs, and AI-driven manipulation tactics, along with reporting and response protocols.
- **Password Hygiene:** Constructing strong, unique passphrases, preventing credential stuffing through non-reuse, and deploying secure password management tools.
- **Two-Factor Authentication (2FA):** Understanding authentication factors, adopting phishing-resistant verification methods, and managing secure account recovery.

- **Safe Use of Devices:** Enforcing prompt software updates, implementing physical screen locks, preventing malware infections, and securing connected smart devices.
- **Safe Use of Public Wi-Fi:** Identifying risks on open networks, avoiding man-in-the-middle attacks, and securing remote connections using Virtual Private Networks (VPNs) and encryption.

Target Audience

This report is designed for all organizational personnel—including full-time employees, remote workers, contractors, and leadership teams. The content assumes no prior technical background, delivering straightforward, non-technical instructions to ensure organization-wide adoption of secure digital habits.

2. Password Hygiene

Section lead: Hasan Peer reviewed by: Iman

2.1 Introduction

Password hygiene constitutes an essential element of cyber security that focuses on the generation, management, and safekeeping of passwords to protect the users' digital accounts from any unauthorized access and use of their personal information. The increasing reliance on digital platforms for communication, study, and financial transactions makes the protection of digital accounts from cybercrimes crucial. Unfavorable password behaviors like creation of weak passwords, repetition of passwords in several accounts, or the storage of the passwords in an unsafe environment can expose the user's accounts to cyber-crimes. According to the National Cyber Security Centre (NCSC, 2024), the use of unique passwords as well as the adoption of security measures is vital in safeguarding one's digital accounts.

2.2 Creating Strong Passwords

Generation of secure passwords represents one of the key components of password hygiene. The insecure password may be cracked by the attacker or subjected to brute-force password attack. Passwords like “password123” and other easy to predict passwords based on the user's birthday, name or sport team represent a great security risk, as the attacker may be able to use these words to crack the password.

The NCSC (2024) suggests combining three random words for the creation of a strong but memorable password. For instance, a combination of random words results in generation of the longer passphrase which does not require memorization of a long string of

characters. The length of the password is important since longer passwords are harder to crack.

It is recommended to avoid the use of personal data and substitution of letters for numbers. It may seem that these techniques will make the password more complex. However, it can lead to generations of predictable passwords. The NIST (2025) warns about strict character composition requirements since the user may generate a predictable password. The user should concentrate on the length, uniqueness, and resistance of the password to guessing attacks.

2.3 Avoiding Password Reuse

Password reuse means utilizing the same password in various accounts. It represents a threat for users because if criminals steal their password on one website, they can use this password to gain access to the rest of their accounts.

So, for example, if criminals obtain some user's password in a breach of the shopping website, they can apply this password to the user's email or bank account. It is called the credential stuffing attack. In case of password reuse, the criminals will be able to hack many accounts of a single person in one incident.

According to NCSC (2024), it is advisable to use a different password in each of your important accounts. It will minimize the consequences of password exposure. Users have to create individual passwords for all their accounts, especially email, bank, and educational accounts. Email accounts must be protected in a special way because they can be used for resetting passwords on other websites.

In case the password was revealed in a data breach or when there is any suspicion about account breach, the user must change his/her password immediately and the passwords in other accounts, where this password was used.

2.4 Using Password Managers

A password manager is a type of software designed to keep user passwords safe. Such programs may even generate new passwords for each account that needs to be protected by an individual.

The use of such software according to NIST (2025) can make the management of user passwords safer because instead of memorizing many passwords, a person can generate

and use unique passwords, which can help to avoid reusing passwords. Besides, some password managers can give warnings about insecure or reused passwords.

While choosing a password manager, one should select a trustworthy program and use a strong password to protect it. Additional protection methods should be used whenever possible, for example, using two-factor or multi-factor authentication. The password should not be reused for any other service because this password grants access to all other user passwords.

At the same time, one should use a password manager correctly. It means that one should protect access to one's device and not share a password vault with anyone else. Password managers are not completely secure, but they may help people maintain good password habits.

2.5 Multi-Factor Authentication

Multi-factor authentication (MFA) refers to an enhanced security protocol that involves providing two or more forms of authentication before logging into an account. This may be anything that the user possesses, anything they know, or anything about the user.

MFA is used to enhance the security of accounts since having just the user password may not be enough for accessing the accounts. For instance, even when the criminal acquires the password, he cannot access the account.

As per the NCSC recommendation, one should apply additional authentication to accounts where possible (NCSC, 2024). Important accounts such as email accounts, financial, and academic accounts should be among those receiving MFA. Authenticating applications and security keys may serve this purpose.

However, MFA does not cover all security vulnerabilities. One needs to secure his password and be careful about the approval of authentication prompts.

2.6 Storing and Protecting Passwords

Passwords need to be secured and not shared needlessly. Storing passwords in viewable sticky notes or in unprotected files and sending them via unprotected messages exposes

the passwords to other parties. The users need to make sure that passwords are not stored in locations where they could be found by unauthorized individuals.

The NCSC (2024) provides that passwords could be stored safely using password managers or appropriate password storage within browsers. Such measures might prove better than using the same password for several accounts.

Additionally, users need to be careful when typing passwords on shared or public computers. They need to refrain from storing credentials on such machines and log out of accounts once done using the accounts. Once the password is inadvertently shared, it needs to be changed immediately.

In addition, it is vital for users to change the default passwords of services or devices whenever they feel necessary. Default credentials might be readily available to others and hence compromise the user's privacy.

2.7 Conclusion

Password hygiene plays an important role in cyber security as passwords provide access to personal information and accounts. The use of weak passwords, password reuse, and poor password storage increases the probability of unauthorized access to personal information. There are certain steps that individuals can take to prevent such threats. Individuals can create long passwords, use password managers, and enable multi-factor authentication for their accounts. Furthermore, users must change passwords once these become compromised. It is crucial to protect passwords from any unauthorized access. Thus, good password hygiene involves responsibility and diligence on the part of the users.

3. Two-Factor and Multi-Factor Authentication

Section lead: Hasan **Peer reviewed by:** Iman

3.1 Introduction

The Two Factor Authentication, otherwise known as 2FA, is an essential cyber security technique whereby a second form of verification is required to access one's digital account or system. The user has to use two forms of authentication instead of just the password. It is necessary since passwords may be stolen, guessed, or exposed through a data breach. The National Cyber Security Centre (n.d.) advises the use of two-step verification for enhanced cyber security. This will help to minimize the risk of unauthorized access despite

password theft. This paper analyses 2FA processes, advantages, common techniques, possible disadvantages, and good practices for enhanced cyber security.

3.2 How Two-Factor Authentication Works

Two-factor authentication implies the need to verify users twice via two distinct authentication methods. The common classification of these factors includes three major categories: something that the user knows, something that the user possesses, and something that the user is. As something the user knows, there is a password or PIN, while something that the user possesses can be a smartphone or a security key. As something that the user is, there is a biometric factor like a fingerprint or facial recognition.

Thus, for instance, when logging into the email account, a user enters a password and confirms his identity in an authentication application. Access is provided only after passing the next step of verification.

The National Institute of Standards and Technology (NIST, 2020) states that to establish that a person is who he claims to be, authentication systems have different forms of evidence. Multi-factor authentication requires factors from separate categories. If the user enters a password and answers the security question, it is not a two-factor authentication because these factors can belong to one category.

3.3 Benefits of Two-Factor Authentication

The first advantage of two-factor authentication is its increased security from unauthorized login. In case a hacker steals a user password using such tactics as phishing, brute-forcing or a data breach, two-factor authentication will prevent him from logging in.

The use of two-factor authentication will be beneficial for the security of significant accounts. Email, banking, college, and work-related accounts may include sensitive information and confidential documents and provide access to other services. Thus, they may benefit from being secured via 2FA since it will lower the negative consequences of having their passwords stolen.

Additionally, using 2FA will allow avoiding risks connected with the password of reuse. In case a password was used on several websites and got compromised, an attacker will have to meet the criteria of the additional verification method as well. The NCSC (n.d.) advises turning on two-factor authentication where it is possible.

But even in this case, 2FA shouldn’t serve as a total alternative to password security.

3.4 Types of Two-Factor Authentication

There are various ways of implementing two-factor authentication, and each of these ways has its security strengths and weaknesses.

The verification code can be sent via text messages to the user's mobile phone. This method is easily accessible and relatively simple. The downside is that it can be hacked by SIM card swapping. Therefore, SMS verification is less secure compared to other types of authentications.

Authentication apps create the time-sensitive verification code or the prompt for approving the request of logging in on the mobile device. Such apps are more secure than SMS verification because their operation is independent of mobile phone network security. The user needs to approve only those login requests that he has initiated.

The hardware security key is an additional physical component that can be plugged in to the computer or connected wirelessly. When the service supports the use of security keys that comply with FIDO2 standards, such keys can offer very good protection against phishing.

Biometrics authentication involves the use of biometric features such as fingerprints or facial recognition. Biometrics could provide convenience to the verification process; however, biometrics are usually used for unlocking the device or authenticating the authentication credential and not the actual second factor in all the systems.

Which method is suitable depends on the nature of the service, requirements of the user, and security needed. The best options where possible include phishing resistant such as hardware security keys or passkeys.

3.5 Limitations and Security Risks

Even though 2FA increases security on the account, there are some drawbacks to its use. Some types of 2FA can be attacked by cyber criminals especially if the user gets deceived and reveals the verification code or approves the malicious request for logging in.

So, for instance, the hacker could keep sending authentication requests to the user to make him click "approve" by mistake. Moreover, the hackers could develop a phishing site where the password and the verification code could be stolen live. So, the user should not reveal the authentication code to anyone and should decline all the unexpected login approval requests.

Also, the drawback of using 2FA is the possible loss of access to the secondary factor of authentication. A user could misplace his phone, damage his security key, or change his phone number.

According to NCSC (n.d.), recovery options should be considered by the users during the process of adding authentication methods. The user is expected to store the recovery codes at a secure place and always update the recovery email addresses and phone numbers. It is important to secure this information since an attacker having access to it can easily bypass normal authentication methods.

3.6 Best Practices for Using Two-Factor Authentication

The user should have 2FA enabled on critical accounts such as emails, bank accounts, and education accounts. This will minimize any chances of an attacker having access to such accounts. In any case, the user should utilize safer ways of protecting themselves from any cyber-attack.

Authentication devices should have screen locks and security upgrades. The user should never share verification codes and login request approvals as they may try to get access to one's accounts.

Besides, backup codes should be kept safe in case of loss of access to the authentication device. It is essential for the user to constantly check the security settings and update account recovery information. This will protect users' accounts from cyber criminals.

3.7 Conclusion

Two-Factor Authentication is one of the crucial security measures designed to prevent digital accounts from unauthorized access. With two-factor authentication in place, hackers have fewer chances of gaining access to the accounts by using compromised login credentials, as 2FA involves the use of two distinct authentication factors. Authentication applications, hardware security keys, and biometric-supported authentication may offer varying degrees of convenience and security to users. Despite the fact that 2FA is far from being a fool-proof security measure, as it is vulnerable to phishing attacks, fraudulent requests for authorization and insecure recovery processes, it is important for users to use 2FA on their important accounts, select secure authentication tools, and ensure account recovery.

4. Phishing and Social Engineering

Section lead: Mackenzie **Peer reviewed by:** Mitchell

4.1 Introduction to Phishing

Phishing is defined as a type of cyberattack implemented via social engineering scam where malicious parties trick people into giving away private data such as card details or personally identifiable information (PII) or installing harmful software. This is often done through the guise of pretending to be a trusted source. “Phishing attacks remain the most prevalent and disruptive type of breach or attack (experienced by 85% of businesses and 86% of charities).” (Rizvi, 2026) is what Rizvi said in the government's annual 2025 cybersecurity review. This shows the current prevalence of phishing attacks in the UK and why it is necessary that awareness and the correct precautions need to be taken.

4.2 Why is phishing so effective?

Phishing’s effectiveness is mainly due to two factors that can work independently but are most effective when paired together. The first of these 2 factors is fraudulently pretending to be a trusted party. The most common way this is done during a phishing attack is manipulating the domain/email, this is the act of manipulating a URL or an outbound email address to resemble a trusted site such as nnicrosoft.com being used to mimic microsoft.com. Simple plays on the way a URL is typed can make an illusion that the site that is asking for your information is legitimate, with some scam sites mimicking the format of the original trusted website. The second of These 2 factors are playing on the psychological factors of victims. As stated in Frontiers in Computer Science, “when people are working under stress, they tend to make decisions without thinking of the possible consequences and options”. (Alkhalil, Z., Hewage, C., Nawaf, L., & Khan, I. (2021). Phishing attacks: Recent comprehensive study and new anatomy. Frontiers in Computer Science, 3, 563060. Doi: HTTPs://doi.org/10.3389/fcomp.2021.563060). Cyber attackers will take advantage of this sort of tactic, often through the guise of creating a false sense of urgency or threat, which can be presented through forms such as faking an account closure based on the emails attached subscribed services, pretending to be a banking agency and asking for account details to resolve an issue, or a delivery company claiming a fraudulent charge and asking for bank details to process a refund. This sort of stress creates an immediate sense of urgency and panic with

unaware individuals which can cause them to not question the scam or take the time to make sure that this situation is a scam, such as missing a small misspelling in an email address sending the scam. Some attacks known as spear-phishing will use a more specialized attack to target specific users which can often have information Used in the scam that can cause the individual to be more concerned. This artificial stress created by these scams make them so dangerous as the weakest point of any cyber security solution is always the end users, as even when measures are put in place, human emotions can be manipulated by malicious attackers a lot easier than a robust system can be breached through other means.

4.3 How to recognize a phishing attack

With the advent of AI, phishing attacks can become more convincing to the end user by using generative AI to mimic company websites, email formats and methods of reaching end users. Certain “fully trained SE models can theoretically learn from each attack and gradually improve their success rate.” ("Digital deception: generative artificial intelligence in social engineering and phishing", 2024, The Artificial Intelligence Review, vol. 57, no. 12, pp. 324.) Which is a cause for concern when thinking about how phishing will affect internet use in the future but is also the reason end users should be made aware of the most common ways to spot these types of attacks as at least one of these indicators should be present in each of these attacks. The first of these is to check the format of the email, some scammers will try to replicate the format of official emails, comparing these to official emails and spotting for differences alongside spelling mistakes could be an instant indicator of a phishing attack. Since the overall knowledge of attackers is limited, they will often try to make sense of urgency using places where a user is signed up. Such as if a delivery company tries to call about a false order, check your account to see if this delivery is in fact there or if a bank contacts you, place separate call to the bank to make sure that this charge is indeed there. The final key indicator is a mistake in the URL or email. In a URL it is often good to check for the lock next to it indicating a secure website as attackers will often purchase cheap domains without any form of encryption as the information that they steal is often not needing to be encrypted as they are not bound to keep your information secure. Emails don't have any form of https that is visible but people within a business with a good email security system will often have an indicator in the email telling them that this email is not from a trusted source and to not reply unless certain. However, the key identifier within

URLs and emails that is unanimous in both is, as mentioned before, subtle misspellings or changes in the URL/email to make the site/email look like the host that it is mimicking. Using these 4 signs, users should be able to identify most phishing attacks sent to them and the best piece of advice when possibly encountering a phishing attack is to step back and look at whether the attack is: sent by a trusted source, the source should be collecting the details, and what the original site says is the case.

4.4 What should you do if you suspect you're a victim of phishing?

If an end user were to suspect that they may have been a target of phishing it is important to first of all stop all contact with the attacker, keeping an attacker in the dark as much as possible when trying to resolve the issue should be the first line of defence against this sort of attack. This is the same for both an individual and in an organization with the one key difference being that end users in an organization This should also report to the relevant IT team. After the communication with the attacker is severed; actions should be undertaken to make sure that any damage from this attack is minimized as much as possible. In the latest 2025 phishing attack trends report, APWG (Phishing activity trends reports (2025) APWG. Available at: https://apwg.org/trendsreports (Accessed: 16 October 2026).) states that 30.9% of all phishing attacks are attacks against online payments and banking with wire transfer attacks increased by 33% compared to previous quarters. The high rate of which bank details are transferred means that, when you have suspected that a phishing attack has taken place; you should ask whether the attack was targeting any banking information or has asked for any form of money to be sent to the attacker, as this can reversed with a phone call to the relevant bank/building society.

5. Safe Use of Devices

Section lead: Iman **Peer reviewed by:** Mitchell

5.1 Introduction

The importance of cyber security is based on the regular use of digital devices, which include computer systems, laptops, smartphones, and tablets for the storage, processing, and usage of important data. Such devices may contain a variety of personal information, including passwords, pictures, documentation, and money-related information, which makes them vulnerable to cyber-crimes. The safe usage of devices includes

implementation of relevant actions to ensure that no unauthorized access or malware infection happens. The National Cyber Security Centre suggests that people should update their devices and software, use secure authentication procedures, and protect devices from malware (NCSC, 2025a). The present report will concentrate on the issue of safe device usage.

5.2 Keeping Devices and Software Updated

One of the most significant considerations for using any electronic device safely is to keep the operating system and software up to date. Manufacturers update devices such as Windows computers, Androids, iPhones, and tablets to resolve any issues, enhance performance, and more importantly to patch security vulnerabilities in the system.

Security vulnerability is a weakness in software that can be used against users by hackers. In case a user is using outdated software on their device, the criminal might try to exploit those vulnerabilities to access their device and steal the data. The NCSC says that security patches are essential in fixing known vulnerabilities which allow hackers to exploit the devices (NCSC, 2025a).

Thus, the users are advised to enable automatic updating where possible and to install the updates that are available to them. It is recommended to update the operating system, browsers, software applications, and security software. The NCSC advises applying the updates immediately when they are available and check whether the auto-updating is working fine (NCSC, 2025a).

5.3 Using Strong Device Passwords and Authentication

The protection of physical accessibility to the device is another aspect of safe device usage. Smartphones, tablets, and computers must be protected by a password, PIN code, fingerprint, or facial recognition. Otherwise, in case somebody has physical access to the device, he/she can view the personal information and access to accounts.

It is recommended to avoid predictable passwords or PIN codes such as “123456”, “000000” or any information that could easily be linked to the user. According to the NCSC, different passwords or PIN codes should be used for different devices and any kind of information such as birthday, name, and other easily guessed details should be avoided (NCSC, 2026).

Two-step verification (2SV) must be enabled if it is supported by the device. This step adds an extra layer of security, which means that even if someone knows your password, it will not help him/her to access the device or corresponding services (NCSC, 2024).

Also, it is recommended to avoid sharing passwords or PIN codes with other people. If one wants to give somebody temporary access to his/her device, he/she should use some other ways instead of sharing the personal password.

5.4 Protection Against Malware

Malware is a malicious software whose role involves performing harmful activities to the device. Malware includes viruses, worms, spyware, and ransomware. Malware has the potential to render a device inoperable, to steal data or damage files, and even prevent users from accessing their information.

The NCSC defines malware as software coming from malicious attachments, downloads, and unauthorized software (NCSC, 2026). This means that safe use of the device entails users being vigilant as to what gets into their devices. Software and application programs are recommended to be downloaded from trusted sources, such as official application stores and websites of the manufacturer.

Another aspect of ensuring a device is safe concerns the issue of keeping security software activated and updated. Modern devices feature built-in malware protection software whose function involves the detection and blocking of malicious software. The NCSC suggests that the malware protection software is activated and regularly updated on the device (NCSC, 2026).

This means that users should not turn off the security software simply because it conflicts with an application or a downloaded program. When the device generates a security warning, users should check the warning first before proceeding further.

5.5 Physical Security of Devices

Using the device safely includes ensuring that the device itself is safe from any harm. Laptops, mobiles, and tablets can be stolen or misplaced, which might result in other people having access to the content stored in those devices. Therefore, users should never leave their devices unattended and unlocked.

Screen locking is a crucial process since it would hinder any person who got hold of the device from accessing the content of it right away. Automatic screen locking should be activated whenever there is no activity on the device for some time.

This type of safety is very important while using laptops and tablets outside the home environment. Users should stay aware of their surroundings to ensure that their device

does not get stolen. NCSC suggests that users take precautionary measures to protect their devices from being stolen (NCSC, 2025b).

5.6 Safe Use of Smart Devices

Appropriate device usage can also apply to internet-enabled smart devices, which may include speakers, security cameras, TVs, thermostats, and other networked home appliances. As such devices can connect to the network, insecure configurations might give the attacker the opportunity to get access to the system illegally.

The NCSC advises changing the default passwords during installation and using secure passwords instead, enabling the two-step verification when available, and switching off remote access when it is not needed (NCSC, 2024).

Smart devices should also be up to date for their entire lifespan. Before buying a new device or continuing to use the old one, people should verify whether its security updates are continued by the manufacturer. This way, the device will not be more vulnerable to attacks because the security holes can no longer be patched.

5.7 Backing Up Important Data

Device safety is also applicable in cases of internet-connected devices. Examples of such devices are smart speakers, security cameras, TVs, thermostats, and other network-enabled home devices. Since they are network enabled, poor configurations might allow an intruder to access them without authorization.

The NCSC encourages users to change passwords during the setup process and use strong passwords. The NCSC also encourages users to enable two-step authentication wherever possible and turn off remote access whenever unnecessary (NCSC, 2024).

Users should ensure that their smart devices receive software updates during their lifetimes. Before purchasing a smart device, one should consider whether its manufacturer is committed to providing security patches. An unsupported device would become more susceptible to threats as any vulnerabilities detected in it may not be patched.

5.8 Conclusion

The safe use of devices is one of the key components of cyber security. Users can substantially decrease their exposure to threats that may arise from using computers, laptops, smartphones, tablets, and other smart devices simply by implementing a number

of security measures. Such measures include the regular updating of operating systems and software, the use of strong passwords and proper authentication, protection against malware, securing devices physically, customization of settings for smart devices and the regular backup of crucial data.

While there is no such thing as a completely secure device, adherence to the aforementioned principles makes the work of cyber criminals much harder as they will find it increasingly hard to exploit popular vulnerabilities. In other words, the safe use of devices relies primarily on the actions of users, which implies that they must be responsible for the maintenance of devices and security updates.

6. Safe Use of Public Wi-Fi

Section lead: Mitchell **Peer reviewed by:** Kenzie

6.1 Introduction to Public Wi-Fi Security

In an increasingly hyper-connected digital landscape, public Wi-Fi networks have transitioned from luxury to a fundamental daily necessity. From coffee shops and airports to public transport and hotels, these networks offer critical convenience, allowing individuals to remain connected to their professional and personal lives while on the move. However, this convenience introduces a significant perimeter of risk for both individuals and the organizations they represent. When users connect to public networks, their data—ranging from routine browsing histories to highly sensitive login credentials and financial information—must travel through an infrastructure that is entirely outside their control. Understanding the evolving technical landscape, the sophisticated methods employed by threat actors, and the psychological factors that drive user behavior is essential for developing a robust cybersecurity posture in public environments.

6.2 The Evolution of Web Encryption and False Trust

Historically, the architecture of the internet was largely unencrypted. Most websites relied on standard HTTP protocols, meaning that any data transferred between a user's device and the host server was transmitted in plain text. Consequently, connecting to a public Wi-Fi network was widely discouraged by security professionals, as malicious actors sharing the same network could effortlessly intercept sensitive information.

Today, the baseline for web security has improved dramatically. The widespread adoption of HTTPS (Hypertext Transfer Protocol Secure) means that the majority of modern websites

utilize Transport Layer Security (TLS) encryption to protect user data in transit. Users are routinely taught to look for the padlock symbol in their browser address bar as an indicator of this encryption. While this makes modern public Wi-Fi inherently safer than it was a decade ago, it has also inadvertently created a false sense of absolute security. Scammers and cybercriminals have adapted to this paradigm by creating their own fraudulent, heavily encrypted websites designed to mimic legitimate corporate or government portals. While a user's data may be securely encrypted while traveling to these malicious sites, the data ultimately lands directly in the hands of the attackers (Federal Trade Commission, 2024). Therefore, the presence of a padlock symbol verifies that the connection is secure, but it does not guarantee that the destination itself is legitimate or trustworthy.

6.3 Human Factors: Psychology, Trust, and Decision Making

Cybersecurity vulnerabilities on public networks are frequently exacerbated by human psychology and behavioral habits. Threat actors actively exploit the implicit trust users place in their physical surroundings. Research indicates that users exhibit "blind-trust" towards networks that are named after the physical locations they are in—for example, automatically trusting an SSID named "Taco Bell's Public Wi-Fi" simply because they are inside the restaurant. Attackers leverage this behavior by setting up "evil twin" access points with these familiar names to lure unsuspecting victims into connecting to malicious infrastructure (Sangeen *et al.*, 2023).

Furthermore, the decision-making process regarding whether to connect to an unsecured network is heavily influenced by immediate emotional and practical needs rather than long-term security considerations. A recent behavioral study highlighted that users consistently prioritize convenience, internet speed, and the preservation of cellular data allowances over the protection of their personal information. Astonishingly, even when individuals possess a clear understanding of the cybersecurity risks associated with public Wi-Fi, they frequently choose to connect anyway. This reckless behavior is largely driven by a psychological phenomenon known as the Fear of Missing Out (FOMO); the emotional urge to remain continuously online and "in the loop" frequently overrides logical security training (Ronchi *et al.*, 2026).

Despite these behavioral challenges, targeted education remains highly effective. Under the framework of the Protection Motivation Theory, when users are explicitly informed about the mechanics of these risks and shown the potential consequences of data theft, they are significantly less likely to reconnect to hostile networks in the future, actively prioritizing their digital safety (Sangeen *et al.*, 2023).

6.4 Technical Threats: Interception and Exploitation

The technical threats lurking on public Wi-Fi are sophisticated and continuously evolving. One of the most prevalent dangers is the Man-in-the-Middle (MitM) attack, where a cybercriminal positions themselves between the user's device and the internet gateway to intercept, read, or alter communications.

Criminals frequently utilize automated, Linux-based hacking frameworks to execute these attacks efficiently. One such prominent tool is Nethercap, which is specifically designed to exploit vulnerabilities in Wi-Fi networks. Nethercap automates complex attack vectors such as Address Resolution Protocol (ARP) spoofing, which maliciously redirects network traffic through the attacker's machine, and Domain Name System (DNS) spoofing, which covertly routes users away from legitimate websites and toward fraudulent clones designed to harvest credentials. Additionally, tools like Nethercap employ packet sniffing to capture sensitive user data right out of the air. The success rate and severity of information theft using these frameworks are exponentially higher when the public network relies on outdated, weak encryption standards such as WEP or WPA (Zulkarnaen *et al.*, 2025).

6.5 Mitigation Strategies and Security Best Practices

To combat the diverse array of threats, present on public Wi-Fi, users and organizations must adopt a layered defense strategy. Technical mitigations at the network level, such as the implementation of the newer, highly secure WPA3 encryption protocol, are vital for defending against automated tools like Nethercap (Zulkarnaen *et al.*, 2025). However, because users cannot control the infrastructure of the public networks they join, personal defensive measures are paramount.

The National Cyber Security Centre (NCSC) explicitly advises that users should treat all public Wi-Fi networks as inherently hostile and unencrypted. Adopting a "Zero Trust" mentality means that users must remain hyper-vigilant and take proactive measures to secure their own devices. Crucially, users should completely avoid performing confidential tasks, such as online banking, accessing proprietary corporate databases, or signing into sensitive accounts while connected to public Wi-Fi. Instead of relying on the network security, the NCSC strongly recommends the use of a Virtual Private Network (VPN). A VPN securely encrypts the entirety of the device's internet traffic—not just the traffic routed to individual HTTPS websites—creating a secure tunnel that shields data from local network eavesdroppers (National Cyber Security Centre, 2026).

Finally, these specific Wi-Fi precautions must be underpinned by general cybersecurity hygiene. Users must deploy strong, complex passwords (utilizing a combination of uppercase, lowercase, numerical, and special characters) and enable Two-Factor Authentication (2FA) across all critical accounts to neutralize the impact of stolen credentials. Keeping security software, operating systems, and internet browsers continuously updated is essential for patching known vulnerabilities before attackers can exploit them. Additionally, users must maintain a healthy skepticism toward unsolicited communications, as scammers will inevitably attempt to breach personal defenses through phishing emails or phone calls, attempting to bypass technical security controls entirely by manipulating the user into handing over their data directly.

7. Conclusion

Section lead: Mitchell **Peer reviewed by:** Kenzie

Cybersecurity is an ongoing operational commitment rather than a one-time technical setup. Across all five domains evaluated in this report, the central finding is clear: technical defenses are most effective when supported by proactive, daily user vigilance.

Key Summary of Findings

- **Phishing Awareness:** Social engineering relies on psychological pressure, false urgency, and domain manipulation. Verifying sender authenticity and stepping back to inspect links before clicking stops the vast majority of email and web scams.
- **Password Hygiene:** Weak and reused passwords remain a primary vulnerability for credential stuffing attacks. Adopting unique multi-word passphrases and using password managers eliminates single points of failure across accounts.
- **Two-Factor Authentication (2FA):** A password alone is no longer adequate for protection. Enabling 2FA adds a necessary second layer of security, blocking unauthorized access even if login credentials are stolen or leaked.
- **Safe Use of Devices:** Outdated software and unmonitored physical hardware invite malware and unauthorized access. Automatic updates, robust screen locks, and updated anti-malware software are required to keep endpoints secure.
- **Safe Use of Public Wi-Fi:** Open public networks expose transmitted data to eavesdropping and man-in-the-middle attacks. Utilizing Virtual Private Networks (VPNs) and delaying sensitive online activities until connected to a trusted network protects unencrypted data.

Closing Action

Building a resilient cybersecurity posture relies on combining technological controls with responsible user behavior. By applying these foundational practices, individuals actively strengthen the organization's overall security culture and significantly reduce the likelihood of a successful cyberattack.

8. References

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- NCSC (2024) – Three random words - Explains how to create strong, memorable passwords and avoid predictable choices.
- NCSC – Password policy: Updating your approach - Covers secure password storage and the use of password management software.
- NIST – Digital Identity Guidelines: Frequently Asked Questions - Explains password length, password managers and password security recommendations.
- National Cyber Security Centre (NCSC) (n.d.) Setting up 2-step verification (2SV). Available at: https://www.ncsc.gov.uk/collection/top-tips-for-staying-secure-online/2-step-verification-2sv (Accessed: 14 October 2026).
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- "Digital deception: generative artificial intelligence in social engineering and phishing", 2024, The Artificial Intelligence Review, vol. 57, no. 12, pp. 324.
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Appendix A: Team Contribution Record

Team member	Sections written	Approx. words	Contribution to intro / conclusion	Sections peer reviewed
Mitchell	6	1108		
Mackenzie	4	1201		
Iman	2,5	1119, 1247		
Hasan	3	1137		

Appendix B: Individual Source Log

Each student: at least **5 credible sources**, including at least **1 peer-reviewed** and **2 authoritative professional or government** sources.

Student	Source (Harvard reference)	Type	Used in section

Mitchell	Federal Trade Commission (2024) Are Public Wi-Fi Networks Safe? What You Need To Know.(Accessed: 2 October 2026)	Government	6
Mitchell	National Cyber Security Centre (NCSC) (2026) 7. Don't trust any network, including your own. Available at: https://www.ncsc.gov.uk/collection/zero-trust/architecture-design-principles/dont-trust-any-network (Accessed: 13 October 2026).	Government	6
Mitchell	Ronchi, C., Neves, C. and Haberli Junior, C. (2026) 'Public Wi-Fi Decisions: Weighing Connectivity Against Privacy Risks', International Journal of Human–Computer Interaction, 42(19), pp. 16792–16811. Available at: https://doi.org/10.1080/10447318.2026.2623215(Accessed: 2 October 2026)	Peer-reviewed	6
Mitchell	Sangeen, M., Bhatti, N.A., Kifayat, K., Alsadhan, A.A. and Wang, H. (2023) 'Blind-trust: Raising awareness of the dangers of using unsecured public Wi-Fi networks', Computer Communications, 209, pp. 359-367.(Accessed: 2 October 2026)	Peer-reviewed	6
Mitchell	Communications, 209, pp. 359-367.(Accessed: 2 October 2026) Zulkarnaen, I., Syarifudin, S. and Rinaldi, S. (2025) 'Wireless Fidelity Network Security Threats (Wi-Fi)', JERIT: Journal of Educational Research and Innovation	Peer-reviewed	6

	Technology, 2(2), pp. 73-82.(Accessed: 13 October 2026)		
Iman	NCSC (2024) – Securing your users' accounts - Covers unique passwords, password managers and two-step verification.	Government	2
Iman	NCSC (2024) – Three random words - Explains how to create strong, memorable passwords and avoid predictable choices.	Government	2
Iman	NCSC – Password policy: Updating your approach - Covers secure password storage and the use of password management software.	Government	2
Iman	NIST – Digital Identity Guidelines: Frequently Asked Questions - Explains password length, password managers and password security recommendations.	Government	2
Iman	National Cyber Security Centre (NCSC) (2024) Smart devices: using them safely in your home. Available at: NCSC Smart Devices Guidance (Accessed: 7 October 2026).	Government	5
Iman	National Cyber Security Centre (NCSC) (2025a) Device security guidance: keeping devices and software up to date. Available at: NCSC Device Security Guidance (Accessed: 7 October 2026).	Government	5

Iman	National Cyber Security Centre (NCSC) (2025b) Advice for end users. Available at: NCSC Advice for End Users (Accessed: 7 October 2026).	Government	5
Iman	National Cyber Security Centre (NCSC) (2026) Cyber Essentials: Requirements for IT Infrastructure v3.3. Available at: Cyber Essentials Requirements (Accessed: 7 October 2026).	Government	5
Iman	National Cyber Security Centre (NCSC) (2026) Small organisations guide to cyber security: Protecting your devices. Available at: Protecting Your Devices (Accessed: 7 October 2026).	Government	5
Hasan	National Cyber Security Centre (NCSC) (n.d.) Setting up 2-step verification (2SV). Available at: https://www.ncsc.gov.uk/collection/top-tips-for-staying-secure-online/2-step-verification-2sv (Accessed: 14 October 2026).	Government	3
Hasan	National Institute of Standards and Technology (NIST) (2020) Digital Identity Guidelines: Authentication and Lifecycle Management. NIST Special Publication 800-63B. Available at: https://pages.nist.gov/800-63-3/sp800-63b.html (Accessed: 14 October 2026).	Government	3
Hasan	National Institute of Standards and Technology (NIST) (n.d.) Digital Identity Guidelines: Frequently Asked	Government	3

	Questions. Available at: https://pages.nist.gov/800-63-FAQ/ (Accessed: 14 October 2026).		
Mackenzie	Rizvi, S. and Fordham, E. (2026) Cyber security breaches survey 2025, GOV.UK. Available at: https://www.gov.uk/government/statistics/cyber-securitybreaches-survey-2025/cyber-security-breaches-survey-2025 (Accessed: 16 October 2026).	Government	4
Mackenzie	Alkhalil, Z., Hewage, C., Nawaf, L., & Khan, I. (2021). Phishing attacks: A recent comprehensive study and a new anatomy. Frontiers in Computer Science, 3, 563060. doi:https://doi.org/10.3389/fcomp.2021.563060	Peer-Reviewed	4
Mackenzie	"Digital deception: generative artificial intelligence in social engineering and phishing", 2024, The Artificial Intelligence Review, vol. 57, no. 12, pp. 324.	Authoritative	4
Mackenzie	Phishing activity trends reports (2025) APWG. Available at: https://apwg.org/trendsreports (Accessed: 16 October 2026).	Government	4

Type: P = peer-reviewed academic · A = authoritative professional / government · O = other credible source

Appendix C: Peer Review Record

Each main section should be checked by another team member before the report is combined.

Check	Yes	Needs attention
Technical explanations are accurate	☐	☐
Important claims have evidence	☐	☐
Definitions are referenced where required	☐	☐
Statistics can be traced to credible sources	☐	☐
Recommendations are supported by evidence	☐	☐
The writing is understandable to a general audience	☐	☐
Citations and references are present	☐	☐

Section	Author	Reviewer	What the reviewer suggested	What was changed
2				
3				
4				
5				
6				

Appendix D: Final Submission Checklist

Before combining your work, agree as a team on numbering, writing style and tone, formatting, technical terms and abbreviations, tables and figures, and Harvard format.

Report structure

- ☐ Title page and contents page (contents updated)
- ☐ Numbered sections and subsections
- ☐ Introduction and conclusion
- ☐ All five main sections present
- ☐ Harvard reference list

Content

- ☐ Technical concepts explained accurately
- ☐ Technical language explained for a general audience
- ☐ Goes beyond simply defining terms
- ☐ Appropriate examples included

☐ Tables and figures used where they help, with captions and sources

Evidence and referencing

☐ Definitions, statistics and recommendations cited

☐ Sources are credible and current

☐ Key statistics traced to original sources

☐ Each student meets the minimum source requirements

☐ Harvard referencing consistent throughout

Team working and tidy-up

☐ Everyone contributed to the introduction and conclusion

☐ Sections peer reviewed and corrections made

☐ Source logs and contribution records completed

☐ All guidance boxes and [placeholders] deleted

☐ GAI declaration completed